Analysis

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pe}v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N}\sum_{j\neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Well-posedness

- Locally Lipschitz, moment-controlled ∇K and $\sigma_x \geq 0$: propagation of chaos on any bounded time interval (McKean, 1967; Bolley et al., 2011) (see also (dW, thesis, 2025))

- Propagation of chaos: $\mu_t^N \to f_t$ solution of the mean-field limit PDE in law as $N \to \infty$ for chaotic moment-controlled initial data

- Quantitatively: $\mathbb{E}\left[d(\mu_t^N, f_t)\right] = O(N^{-1/2})$

Analysis

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pe}\,v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N}\sum_{j\neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Well-posedness
  - Locally Lipschitz, moment-controlled ∇K and $\sigma_x \geq 0$: propagation of chaos on any bounded time interval (McKean, 1967; Bolley et al., 2011) (see also (dW, thesis, 2025))
  - Propagation of chaos: $\mu_t^N \to f_t$ solution of the mean-field limit PDE in law as $N \to \infty$ for chaotic moment-controlled initial data
    - Quantitatively: $\mathbb{E}\left[d(\mu_t^N, f_t)\right] = O(N^{-1/2})$

Analysis

$$\begin{cases} \mathrm{d}X_t^i & = \mathrm{Pe}\,v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i & = \chi n(\theta_t^i) \cdot \frac{1}{N}\sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Well-posedness
  - Singular (Newtonian) ∇K: $|\nabla K(x)| \sim |x|^{-1}$
    - For $\sigma_x > 0$, **if τ small enough**, solutions exist for all N and any time $t \geq 0$
    - And: convergence in law to mean-field limit as $N \to \infty$ ($L^1 \cap L^\infty$ chaotic initial law)
  - Via recent compactness method for Keller-Segel (Jourdain & Fournier, Ann. Appl. Prob., 2017) (see dW, thesis, 2025)